Mystery Meat Analysis

BIOLM0051

Archie Benn

Scenario

A suitcase containing several packages of unidentified frozen meat has been intercepted at Heathrow Airport by UK Border Force officers during routine inspections. The shipment had no import documentation, raising concerns that the meat may originate from protected or endangered species.

Given the serious implications for public health, food safety, and wildlife conservation, the case has been referred to the Animal and Plant Health Agency (APHA). As a bioinformatician assisting with the inquiry, you have been tasked with determining the species of origin for each meat sample. DNA barcoding has already been performed in the laboratory, and you have now received the resulting FASTAQ sequence files for analysis.

Abstract

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Introduction

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Methods

Pipeline Overview

Some python code:

Results

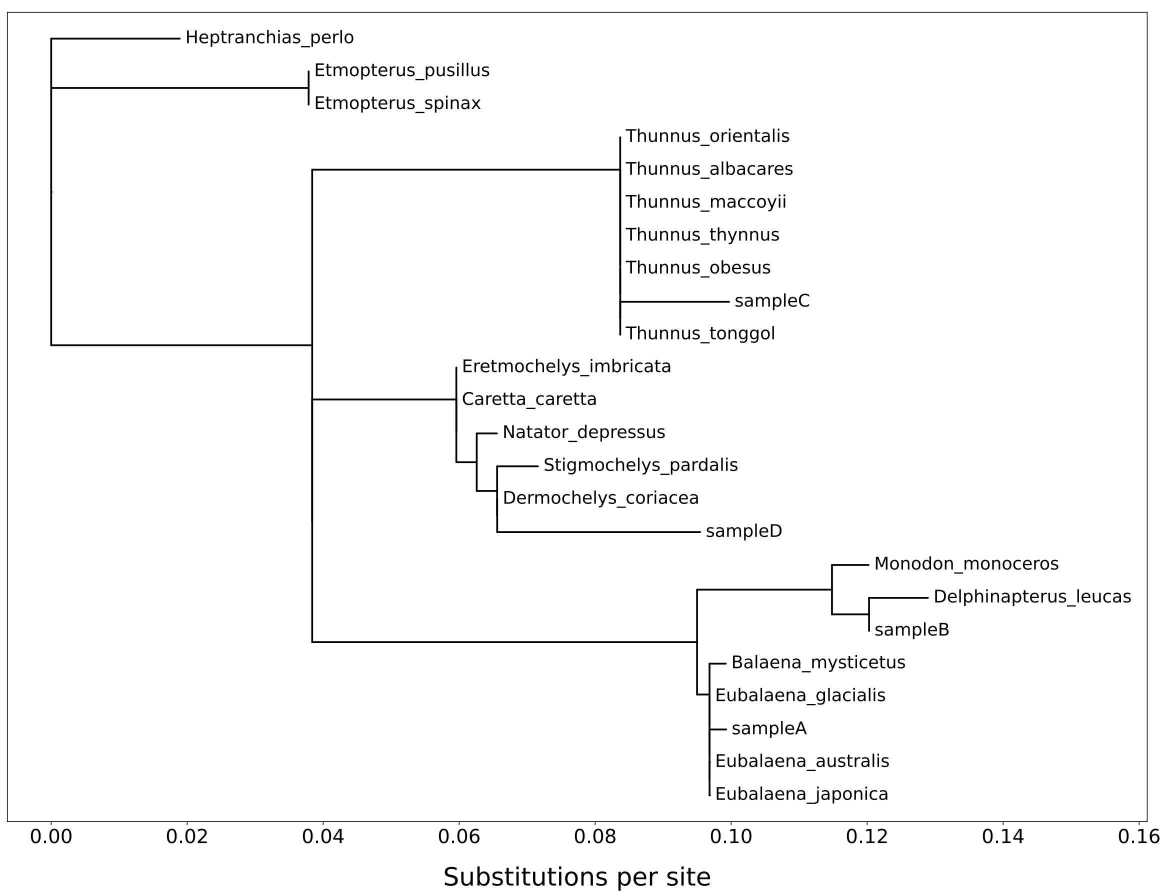

Figure 1: Phylogenetic placement of unknown meat samples and their top BLAST hits

Discussion

`sampleD_part1` nt length is 524bp, whereas the efetch FASTA lengths are 525bp. This single deletion could be causing the frame shift visible towards the end of the `GAINFITTIIN` region in Fig. 2. This deletion may have been caused by an error during sequencing.

Figure X. Maximum-likelihood tree from the original protein alignment. Figure Y. Maximum-likelihood tree from a manually corrected alignment in which one ambiguous base - N - was inserted into `sampleD_part1` at the suspected deletion site to restore the reading frame. This tree is shown to assess the impact of the putative sequencing error on branch length and topology.

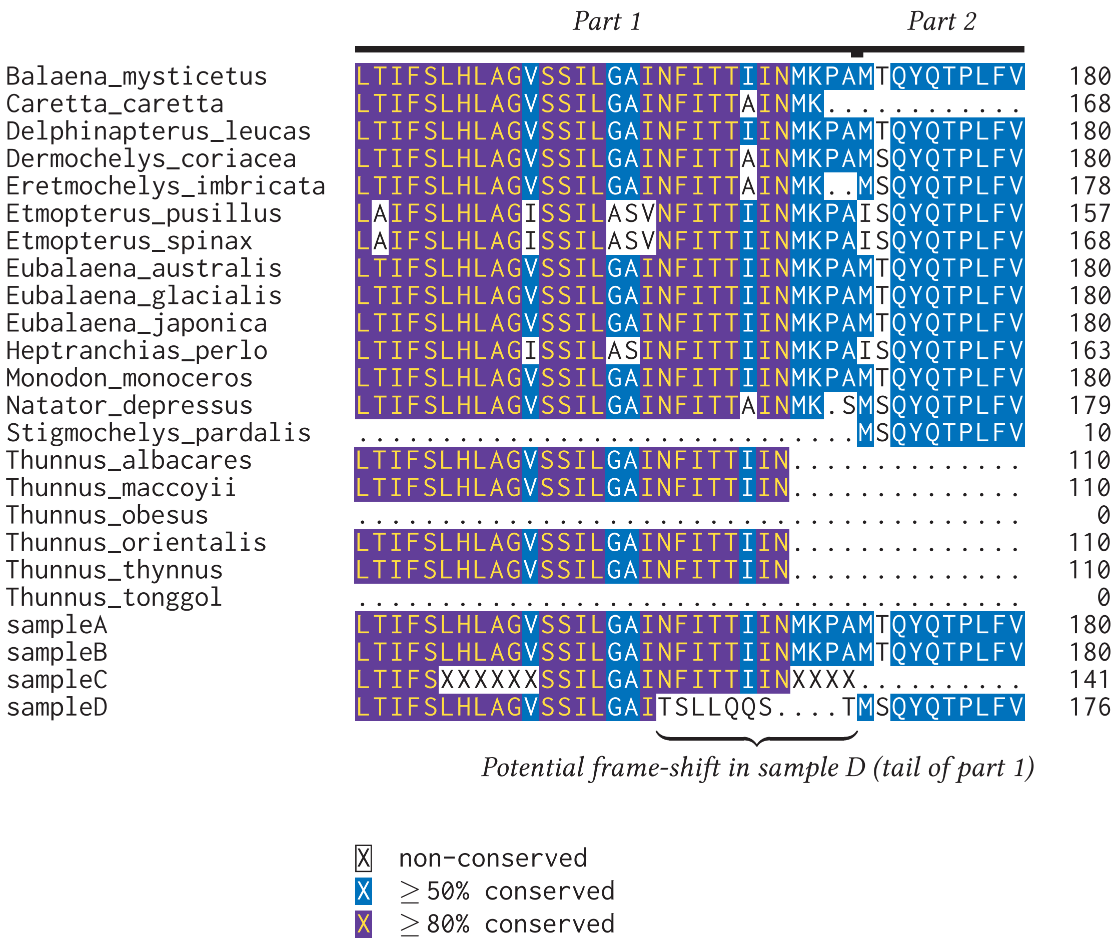

Figure 2: Subset of supermatrix alignment with potential frame shift error in `sampleD` near the highly-conserved `INFITTIIN` region.

```
Dermochelys_c     441  CACCTAGCTGGTGTTTCATCAATTTTAGGAGCTATTAACTTCATTACTAC     490
                       |||||||||||||||||||||||||||||||||| |||||||||||||
sampleD_part1     451  CACCTAGCTGGTGTTTCATCAATTTTAGGAGCTATT-ACTTCATTACTAC     499
```

Figure 3: Nucleotide Needleman-Wunsch pairwise global alignment of the tail end of `sampleD_part1` against closest reference, *Dermochelys coriacea*, and showing a possible sequencing base miss at position 486 in `sampleD_part1`.

References

Supplementary Data

Supplementary data and all scripts can be found within the GitHub repository for this project: https://github.com/archiebenn/BIOLM0051_analysis.git

Appendix: Figures for Manual Correction of Suspected Sequencing Error

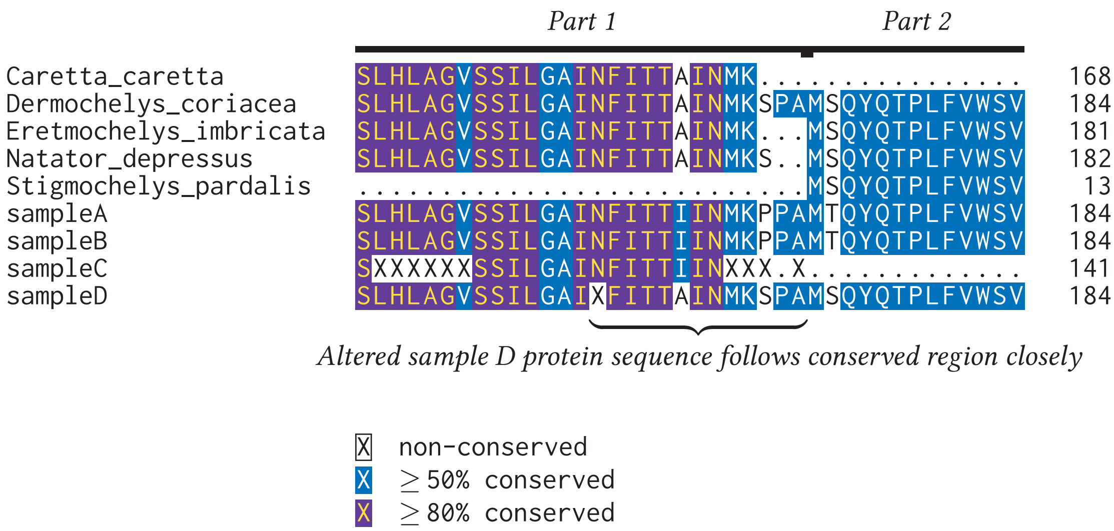

Figure 4: Subset of supermatrix alignment of `sampleD` with clade references and other samples resulting from manually edited `sampleD` nucleotide sequence (other references left out of figure contribute to the conserved residue %). The altered protein sequence from one ambiguous base addition in the nucleotide sequence aligns much more closely to highly-conserved `INFITTIIN` region seen in other sequences.

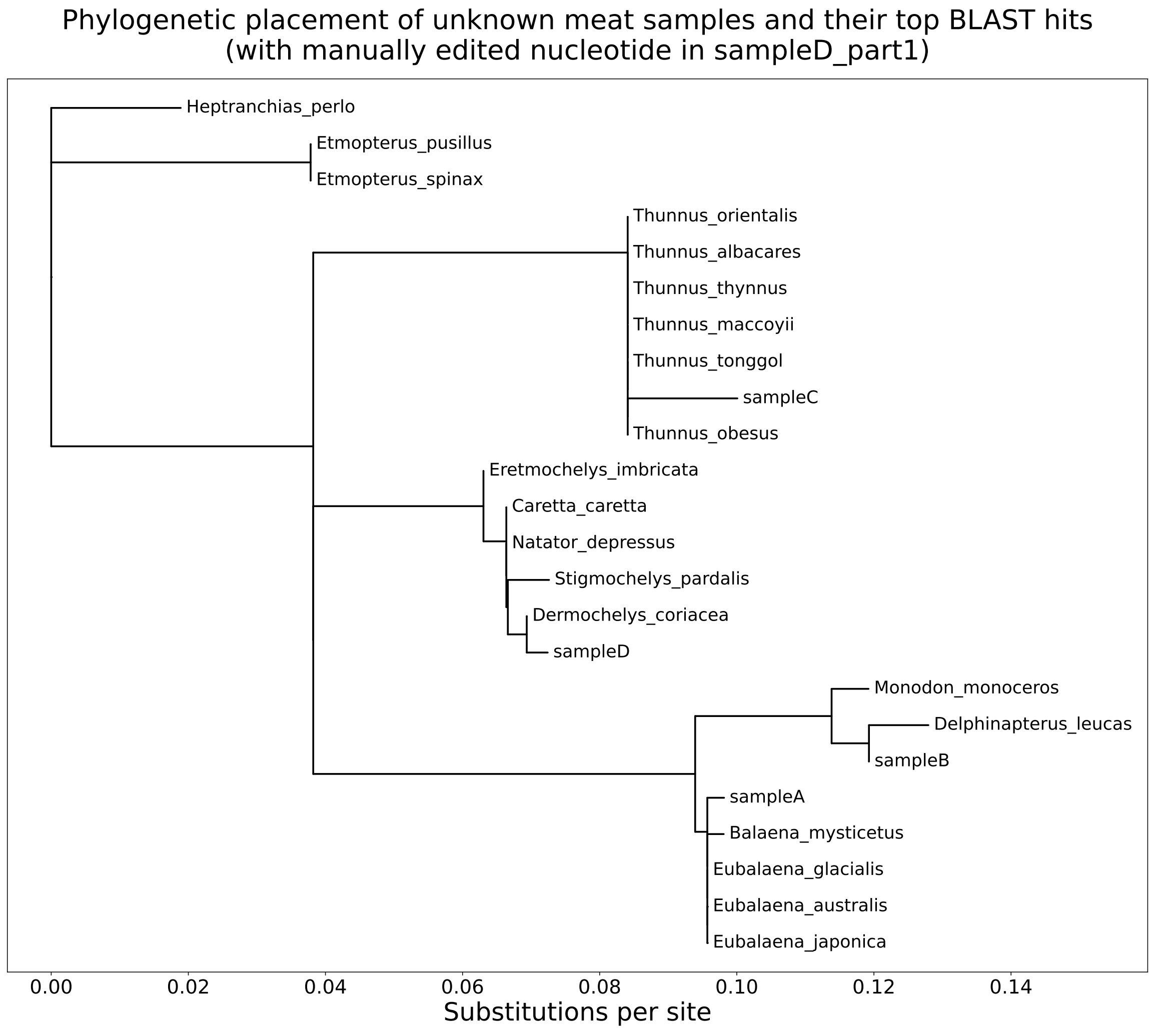

Figure 5: Phylogeny of Mystery Meat samples and closest BLAST matches.